lined by cuboidal epithelium.The follicles are filled with colloid material called **thyroglobulin.**

An amino acid **tyrosine** and **iodine** are involved in the formation of thyroid hormone. The hormones secreted by the thyroid gland are

a. Triiodothyronine (T_3)
b. Tetraiodothyronine or Thyroxine (T_4)

Functions of thyroid hormones

The functions of thyroid hormones are

- Production of energy by maintaining the **Basal Metabolic Rate** (BMR) of the body.
- Helps to maintain normal body temperature.
- Influences the activity of central nervous system.
- Controls growth of the body and bone formation.
- Essential for normal physical, mental and personality development .
- It is also known as **personality hormone**.
- Regulates cell metabolism.

More to Know

Edward C. Kendal in 1914 first crystallised thyroxine hormone. Charles Harrington and George Barger identified the molecular structure of thyroxine in 1927. Thyroid gland requires "120 µg" of iodine everyday for the production of thyroxine.

Thyroid Dysfunction

When the thyroid gland fails to secrete the normal level of hormones, the condition is called **thyroid dysfunction**. It leads to the following conditions

Hypothyroidism

It is caused due to the decreased secretion of the thyroid hormones. The abnormal conditions are simple goitre, cretinism and myxoedema.

Goitre



It is caused due to the inadequate supply of iodine in our diet. This is commonly prevalent in Himalayan regions due to low level of iodine content in the soil. It leads to the enlargement of thyroid gland which protrudes as a marked swelling in the neck and is called as **goitre.**

Cretinism

It is caused due to decreased secretion of the thyroid hormones in children. The conditions are stunted growth, mental defect, lack of skeletal development and deformed bones. They are called as **cretins**.

Myxoedema

It is caused by deficiency of thyroid hormones in adults. They are mentally sluggish, increase in body weight, puffiness of the face and hand, oedematous appearance.

Hyperthyroidism

It is caused due to the excess secretion of the thyroid hormones which leads to Grave's disease. The symptoms are protrusion of the eyeballs (Exopthalmia), increased metabolic rate, high body temperature, profuse sweating, loss of body weight and nervousness.

16.2.3 Parathyroid Gland

The parathyroid glands are four small oval bodies that are situated on the posterior surface of the thyroid lobes. The **chief cells** of the gland are mainly concerned with secretion of **parathormone**.

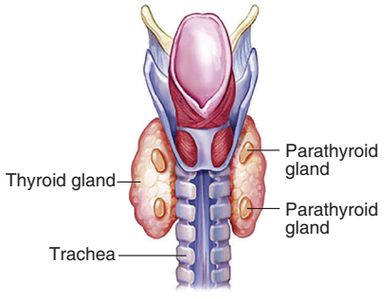


Figure 16.9 Parathyroid Gland

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